CS224V: Agentic AI

Reading List
Fall 2026

1 Large Language Models (LLMs) Introduction

1.1 Pretraining LLMs

1. Attention: the classic transformer paper (Vaswani et al., 2017) [Mandatory]
2. GPT-3 (Brown et al., 2020)
3. Instruct-GPT (Ouyang et al., 2022)
4. LLAMA (Touvron et al., 2023)
5. Deepseek-R1: Reasoning LLMs via reinforcement learning (DeepSeek-AI et al., 2025).

1.2 Prompting LLMs

1. Chain-of-thought (Wei et al., 2023)[Mandatory]
2. Self-consistency (Wang et al., 2023)
3. The Prompt Report: A Systematic Survey of Prompting Techniques (Schulhoff et al., 2024)

1.3 Tools

1. MCPVerse: a benchmark and comparison of commercial LLMs for agentic tool use (Lei et al., 2025).
2. Cursor: a popular AI agent (Cursor).

1.4 Distillation

1. OPSD: On-policy self-distillation (Zhao et al., 2026)
2. S1: Simple test-time scaling (Muennighoff et al., 2025)

2 Knowledge Curation

1. STORM: writing Wikipedia-like articles from scratch (Shao et al., 2024) [Mandatory]
2. Co-STORM: collaborative STORM (Jiang et al., 2024) [Mandatory]
3. DataSTORM: Deep Research on Large-Scale Databases using Exploratory Data Analysis and Data Storytelling (Liu et al., 2026) [Mandatory]

3 Question-Answering on Free Text

3.1 Neural Information Retrieval Models

1. ColBERT (Khattab and Zaharia, 2020) [Mandatory]
2. M3-Embedding (Chen et al., 2024)
3. RankGPT (Sun et al., 2023)

3.2 Retrieval + Generation

1. WikiChat: nearly hallucination-free Q&A grounded on Wikipedia. (Best Research of the Year, Wikimedia, 2024).(Semnani et al., 2023)
2. Citation generation (Gao et al., 2023)
3. Active retrieval augmented generation (Jiang et al., 2023)

3.3 Over a Document Set

1. SLIDERS: Documents to structured data with reconciliation (Joshi et al., 2026) [Mandatory]
2. DocETL: Map and Reduce over documents (Shankar et al., 2024)

3.4 Evaluation

1. Evaluating Verifiability in Generative Search Engines (Liu et al., 2023a)
2. Generating Benchmarks for Factuality Evaluation (Muhlgay et al., 2023)
3. Evaluating Open-Domain Question Answering in the Era of Large Language Models (Kamalloo et al., 2023)

4 Question Answering on Databases, Knowledge Graphs, and Heterogeneous Sources

1. SUQL: Structured and Unstructured Query Language (Liu et al., 2024b) [Mandatory]
2. Spinach: SPARQL-based Agent using an Agentic Approach (Liu et al., 2024a) [Mandatory]
3. Spider 2.0: Spider 2.0: Hardest Text-to-SQL Dataset—Real-World Enterprise Workflow (Cao et al., 2024)
4. ReFinED: Named Entity Disambiguation (NED) (Ayoola et al., 2022)
5. GraphReader: Builds a graph from text and answers questions by navigating the graph (Li et al., 2024)
6. GraphRAG (Edge et al., 2024)
7. Biomni: A General-Purpose Biomedical AI Agent (Huang et al., 2025a)

5 Conversational Agents

1. Genie Worksheet–non-hallucinating conversational agent framework (Joshi et al., 2025). [Mandatory]
2. Zero-shot Persuasive Chatbots with LLM-Generated Strategies and Information Retrieval (Furumai et al., 2024)

6 Long Context

1. FinanceBench dataset fo answering questions of long documents (Islam et al., 2023).
2. Multi-doc question-answering: a benchmark for long-context models (Huang et al., 2025b).

7 Formal Representations of Knowledge

1. LEMONADE: multilingual abstractive event dataset (Semnani et al., 2025).
2. An overview of Knowledge Graphs: (Hogan et al., 2021)
3. YAGO3: Curating a knowledge graph from Wikipedia (Mahdisoltani et al., 2013)
4. Biomedical Knowledge Graph Curation: (Percha and Altman, 2018)

8 Formal Reasoning - Theorem Proving

1. SatIR: Scalable High-Recall Constraint-Satisfaction-Based Information Retrieval for Clinical Trials Matching (Zhou et al., 2026b) [Mandatory]
2. Accountable AI with Grounded, Faithful, Consistent, Actionable Rationales: A Case Study in Clinical Trial Matching with VERDICT (Zhou et al., 2026a) [Mandatory]
3. SATLM: Translate NL into SAT (satisfiability) formulae and solve with theorem proving. (Ye et al., 2024)
4. Logic-lm: Empowering large language models with symbolic solvers for faithful logical reasoning (Pan et al., 2023)
5. Lang2ltl: Translating natural language commands to temporal specification with large language models (Liu et al., 2022)

9 Meta Agents and Meta Optimization

1. ADAS: Automated design of agentic systems—a meta agent that programs new agents in code (Hu et al., 2025) [Mandatory]
2. AFlow: Automating agentic workflow generation with Monte Carlo tree search (Zhang et al., 2025)
3. DSPy: Compiling declarative language model calls into self-improving pipelines (Khattab et al., 2024)
4. OPRO: Large language models as optimizers (Yang et al., 2024)
5. TextGrad: Automatic "differentiation" via text (Yuksekgonul et al., 2024)
6. Gepa: Reflective prompt evolution can outperform reinforcement learning (Agrawal et al., 2026)
7. AlphaEvolve: A coding agent for scientific and algorithmic discovery (Novikov et al., 2025)
8. Self-taught optimizer (stop): Recursively self-improving code generation (Zelikman et al., 2023)
9. Darwin Gödel machine: open-ended evolution of self-improving agents (Zhang et al., 2026a)

10. Hyperagents (Zhang et al., 2026b)

11. Dream-RSI: Recursive Self-Improvement through Evolving Worlds (Zheng et al., 2026)

10 Evaluation of LLMs

1. HELM (Liang et al., 2022),

2. Repairing the Cracked Foundation: A Survey of Obstacles in Evaluation Practices for Generated Text (Gehrmann et al., 2022)

3. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena (Zheng et al., 2023)

4. G-Eval: NLG Evaluation using GPT-4 with Better Human Alignment (Liu et al., 2023b)

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